

Modern Digital Banking Dashboard

Project Statement:

Unifies accounts, transactions, budgeting, bills, and rewards into a single personal banking hub with insights and alerts.

Key Features:

- Multi-account overview & transaction feeds
- Expense categorization & monthly budgets
- Bill payment reminders & workflows
- Rewards tracking & insights
- Currency conversion summaries

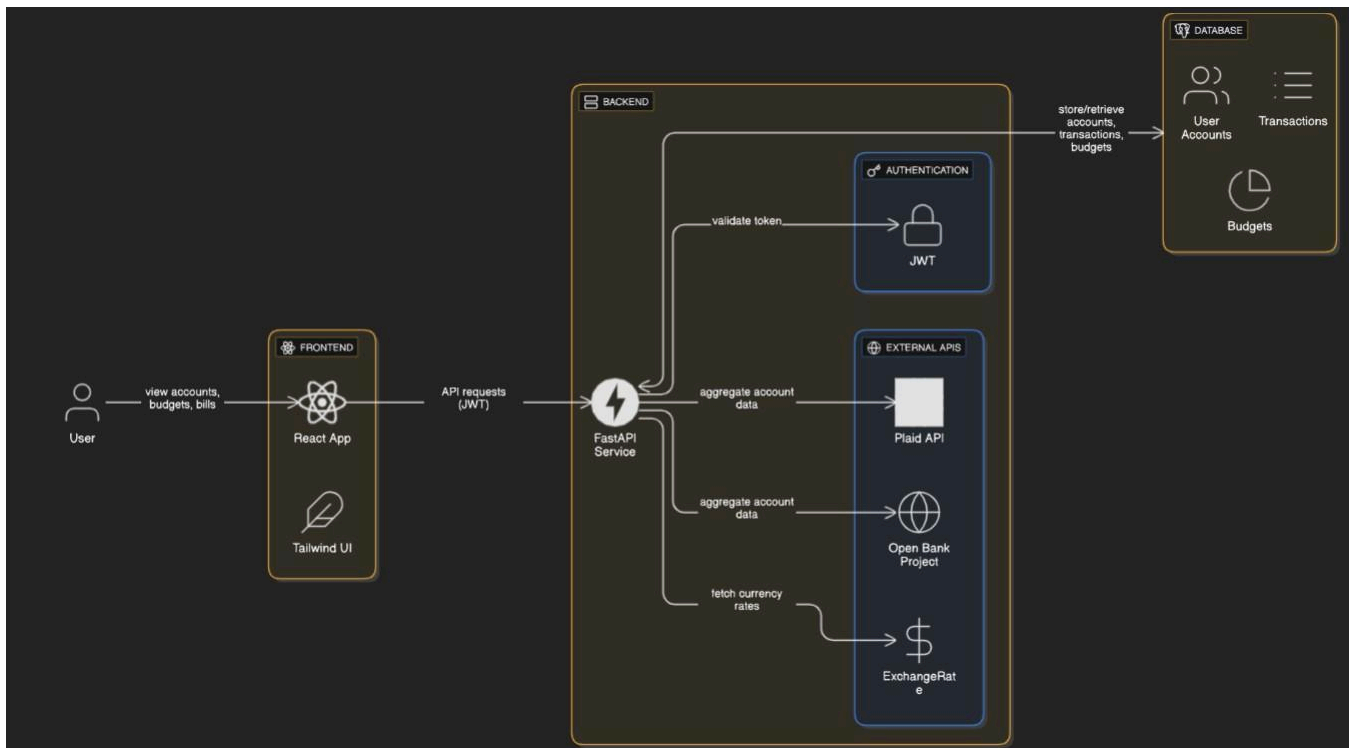
Tech Stack:

- Frontend: React.js + Tailwind CSS
- Backend: FastAPI
- Database: PostgreSQL
- Authentication: JWT (access + refresh)
- **Extras:** Webhooks/polling for account refresh; Integrations (Plaid/Open Bank Project, ExchangeRate API)

Modules:

- Module A: Auth & User Setup
- Module B: Accounts & Transactions
- Module C: Budgeting & Categorization
- Module D: Bills, Reminders & Rewards
- Module E: Insights, Alerts & Admin

Architecture Diagram:



8-Week Milestone Plan

Milestone 1: Weeks 1–2 – Foundations & Accounts

Week 1: Auth + base schema; Accounts CRUD (manual link for prototype)

Week 2: Transaction ingestion (CSV import first), list & details UI

Expected Output:

Accounts + transactions visible in dashboard.

Milestone 2: Weeks 3–4 – Categorization & Budgets

Week 3: Category rules (merchant/keywords), manual recategorization

Week 4: Monthly budget creation & progress bars

Expected Output:

Budget status and categorized spend charts.

Milestone 3: Weeks 5–6 – Bills & Rewards

Week 5: Bills CRUD, reminders (Celery emails/SMS)

Week 6: Rewards tracking (points balance), currency summaries via ExchangeRate API

Expected Output:

Active bill reminders + rewards dashboard.

Milestone 4: Weeks 7–8 – Insights & Alerts

Week 7: Insights (cash flow, top merchants, burn rate), Alerts table

Week 8: CSV/PDF exports, QA, deployment

Expected Output:

Insight widgets, alert center, exportable reports, and production build.

Expected Project Outcome:

By Week 8, users manage multi-account finances, budgets, bills, and rewards with insights and alerts in a polished app.

Database Schema:

- **Users:** id (INT, PK), name (VARCHAR), email (VARCHAR, UNIQUE), password (VARCHAR), phone (VARCHAR), kyc_status (ENUM: 'unverified', 'verified'), created_at (TIMESTAMP)
- **Accounts:** id (INT, PK), user_id (FK to Users.id), bank_name (VARCHAR), account_type (ENUM: 'savings', 'checking', 'credit_card', 'loan', 'investment'), masked_account (VARCHAR), currency (CHAR(3)), balance (NUMERIC), created_at (TIMESTAMP)
- **Transactions:** id (INT, PK), account_id (FK to Accounts.id), description (VARCHAR), category (VARCHAR), amount (NUMERIC), currency (CHAR(3)), txn_type (ENUM: 'debit', 'credit'), merchant (VARCHAR), txn_date (TIMESTAMP), posted_date (TIMESTAMP)
- **Budgets:** id (INT, PK), user_id (FK to Users.id), month (INT), year (INT), category (VARCHAR), limit_amount (NUMERIC), spent_amount (NUMERIC), created_at (TIMESTAMP)
- **Bills:** id (INT, PK), user_id (FK to Users.id), biller_name (VARCHAR), due_date (DATE), amount_due (NUMERIC), status (ENUM: 'upcoming', 'paid', 'overdue'), auto_pay (BOOLEAN), created_at (TIMESTAMP)
- **Rewards:** id (INT, PK), user_id (FK to Users.id), program_name (VARCHAR), points_balance (INT), last_updated (TIMESTAMP)
- **Alerts:** id (INT, PK), user_id (FK to Users.id), type (ENUM: 'low_balance', 'bill_due', 'budget_exceeded'), message (TEXT), created_at (TIMESTAMP)
- **AdminLogs:** id (INT, PK), admin_id (FK to Users.id), action (TEXT), target_type (VARCHAR), target_id (INT), timestamp (TIMESTAMP)

